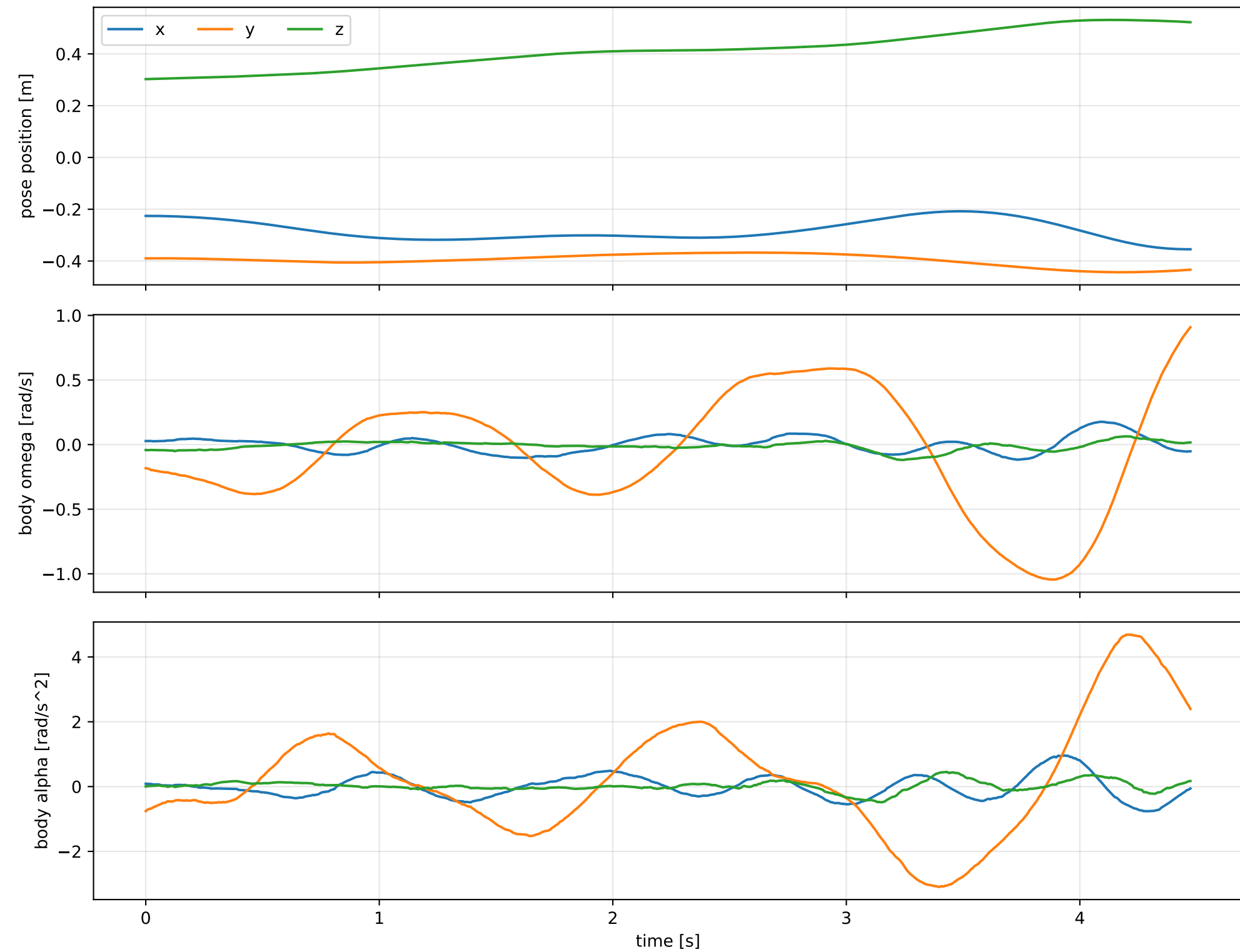
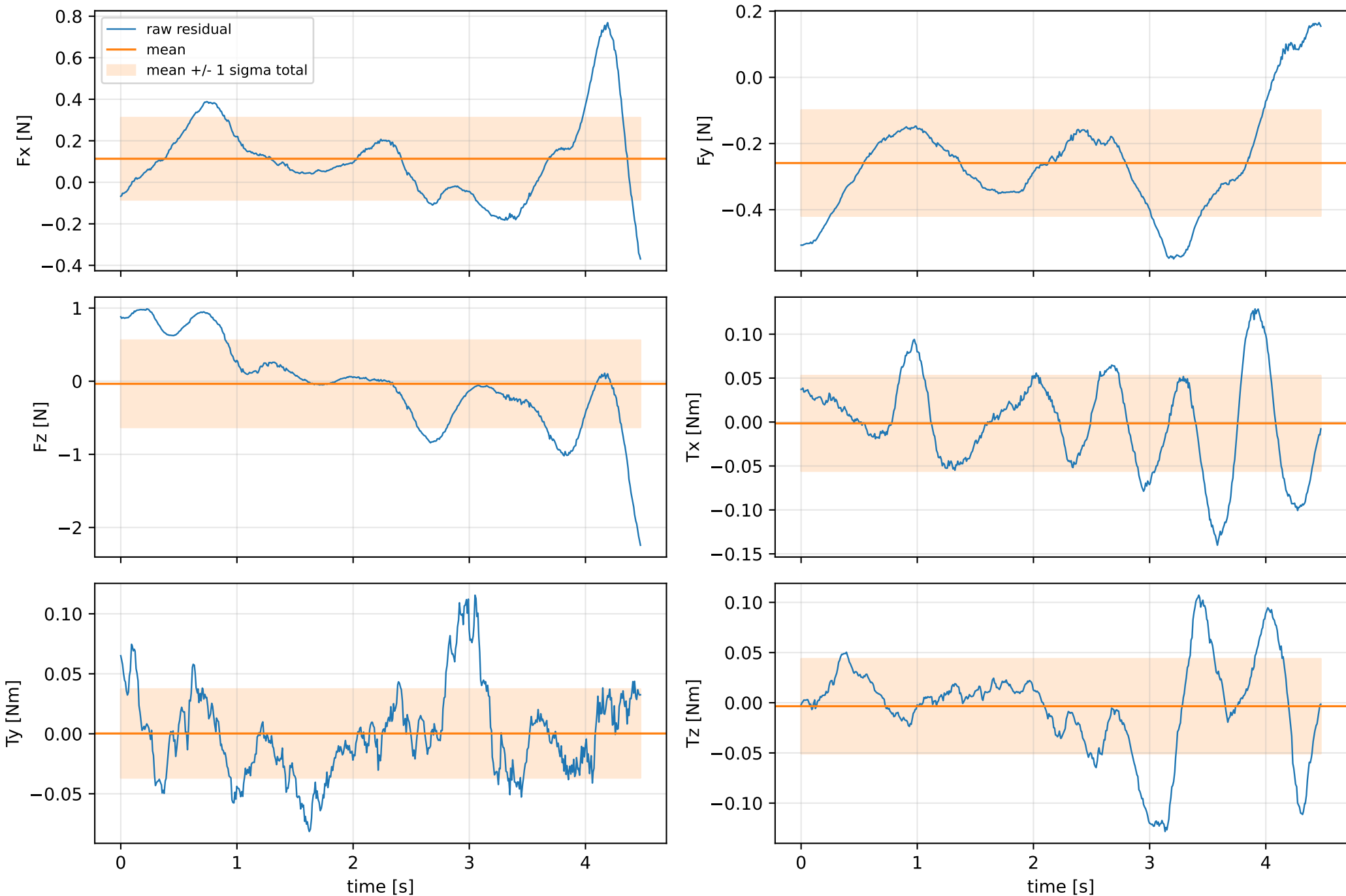


cog\_z\_nominal: SG trajectory and derived motion



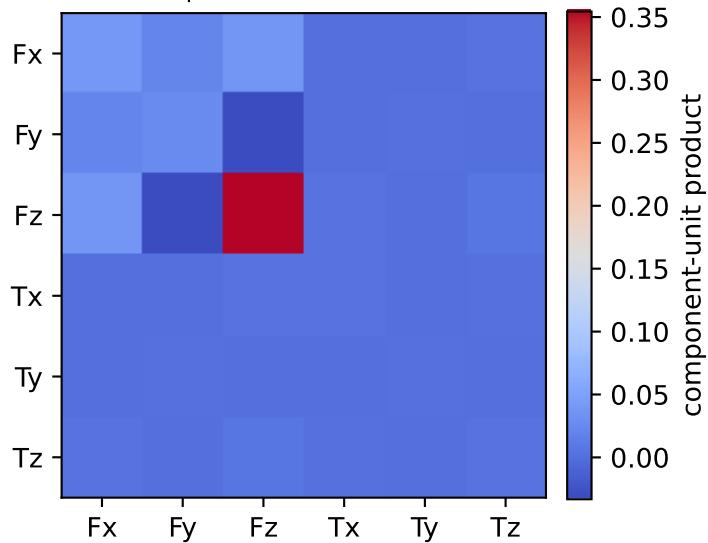
cog\_z\_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)  
mass gauge fixed to nominal mass = 2.35159759 kg



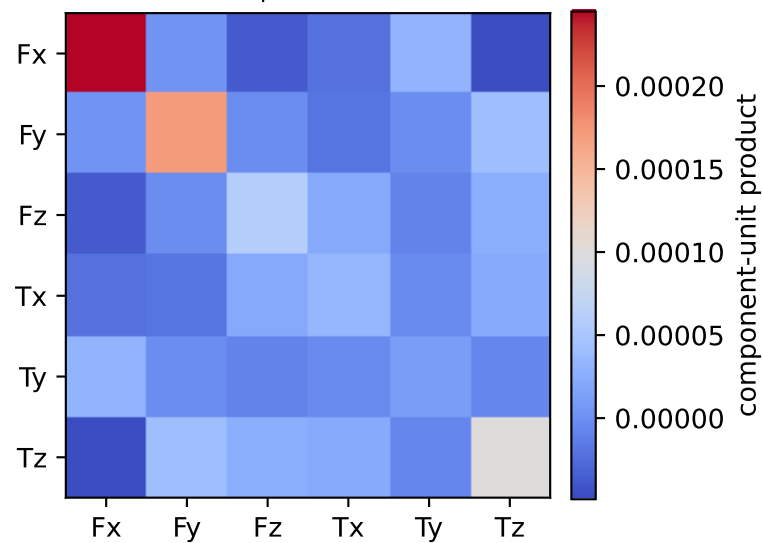
vector RMS about zero: force 0.70672 N, torque 0.080866 Nm; component std about mean: force [0.1977 0.1599 0.5953], torque [0.0542 0.037 0.0472]

cog\_z\_nominal: residual-wrench covariance decomposition in nominal-mass gauge  
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.004 0.005 0.052 0.361]

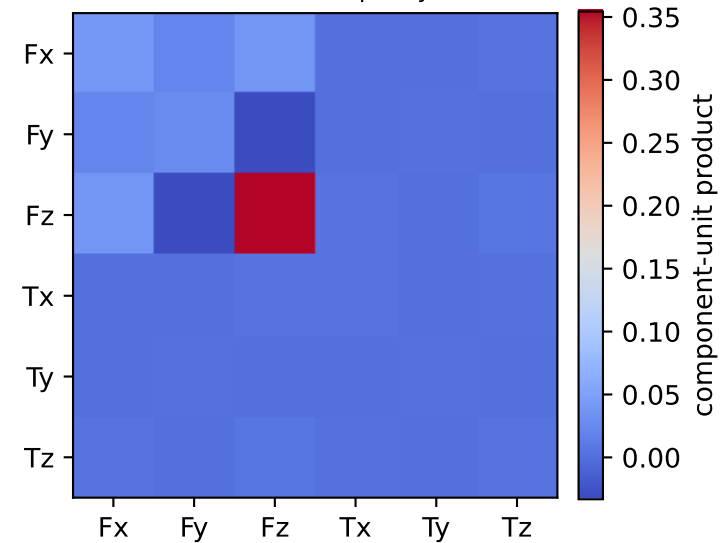
empirical total covariance



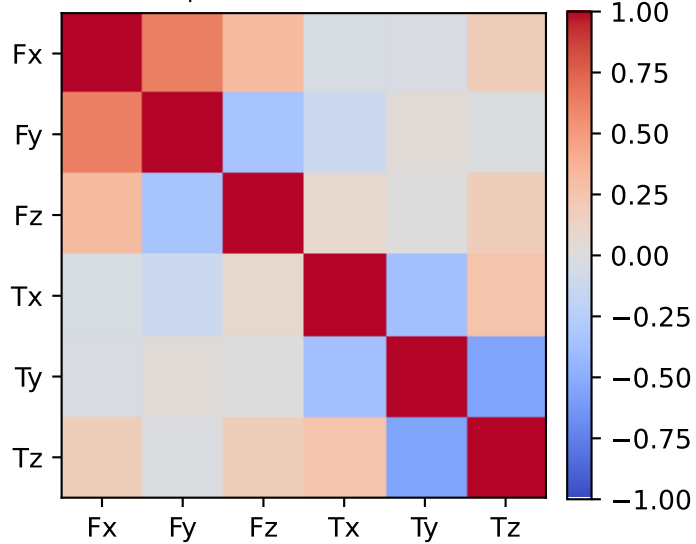
mean full-SG prediction covariance



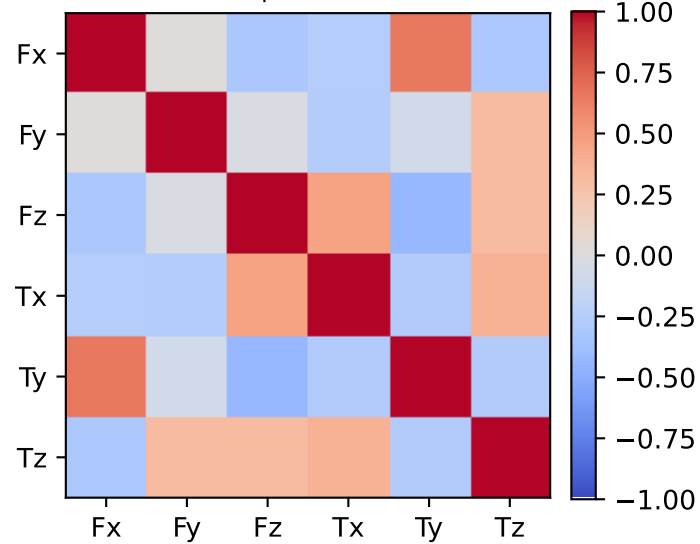
PSD excess model discrepancy covariance



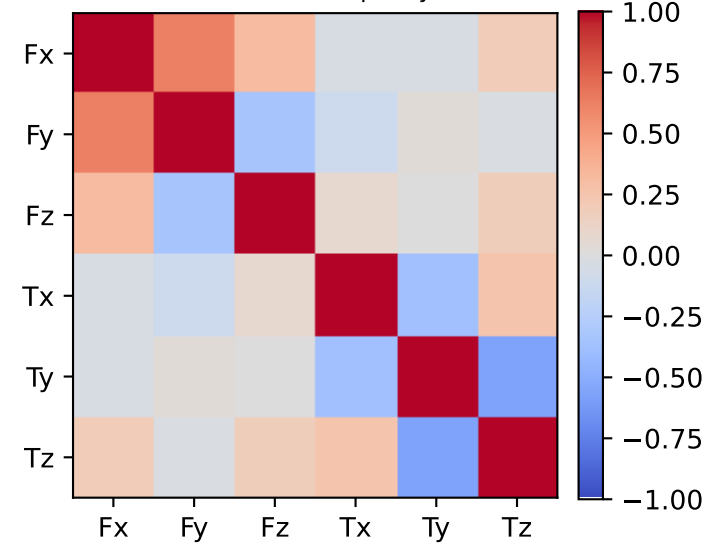
empirical total correlation



mean full-SG prediction correlation

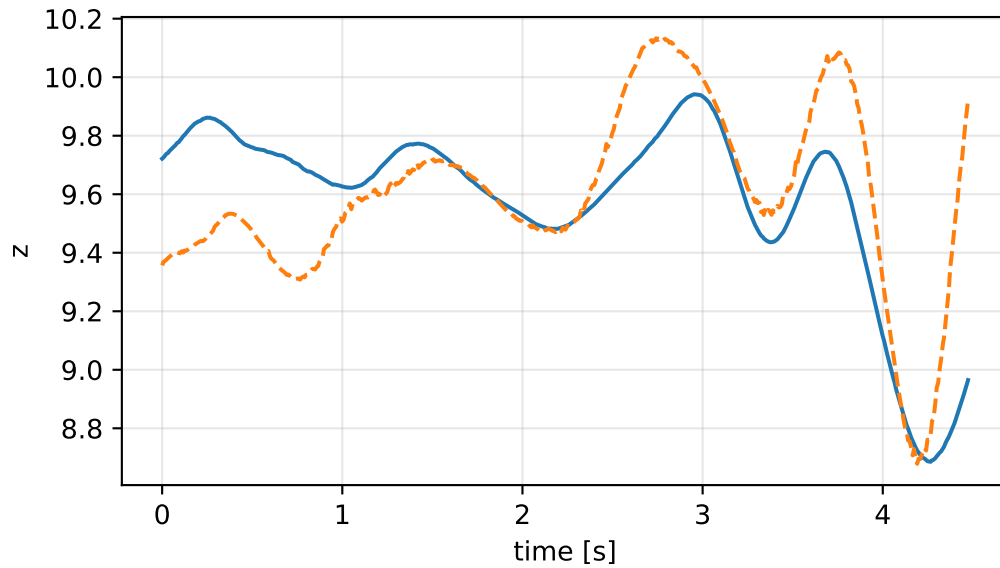
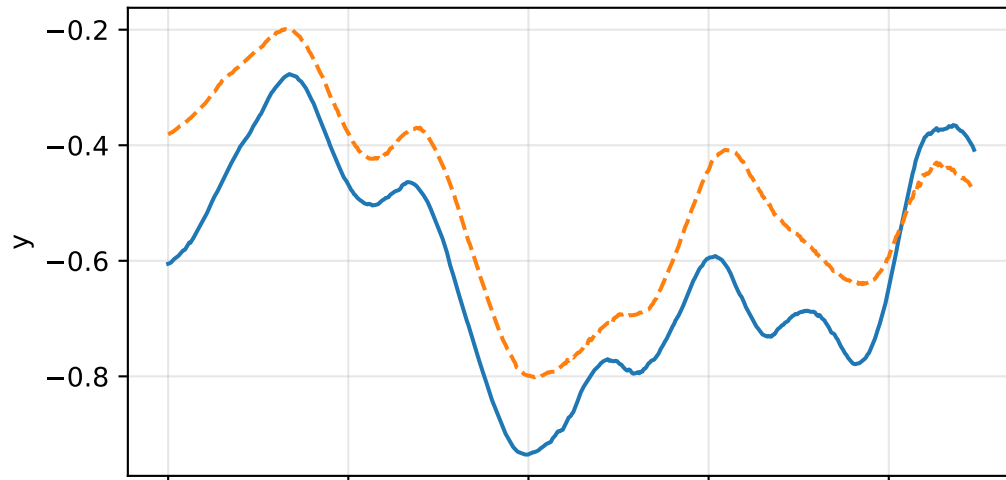
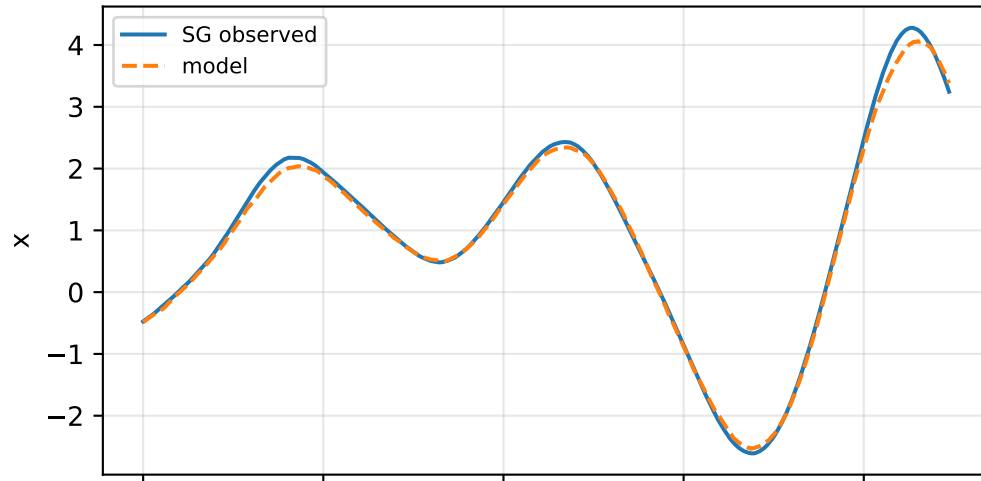


PSD excess model discrepancy correlation

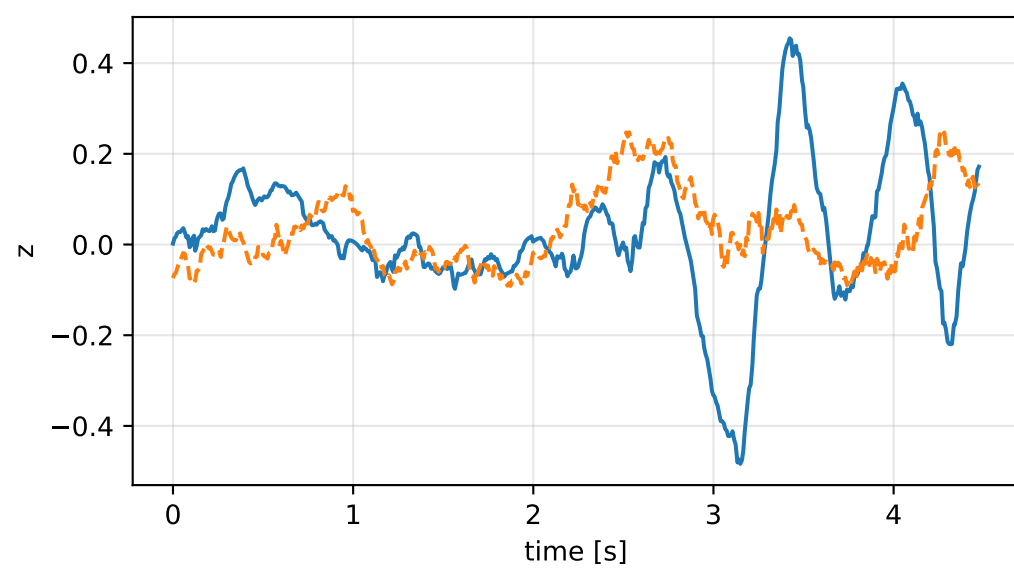
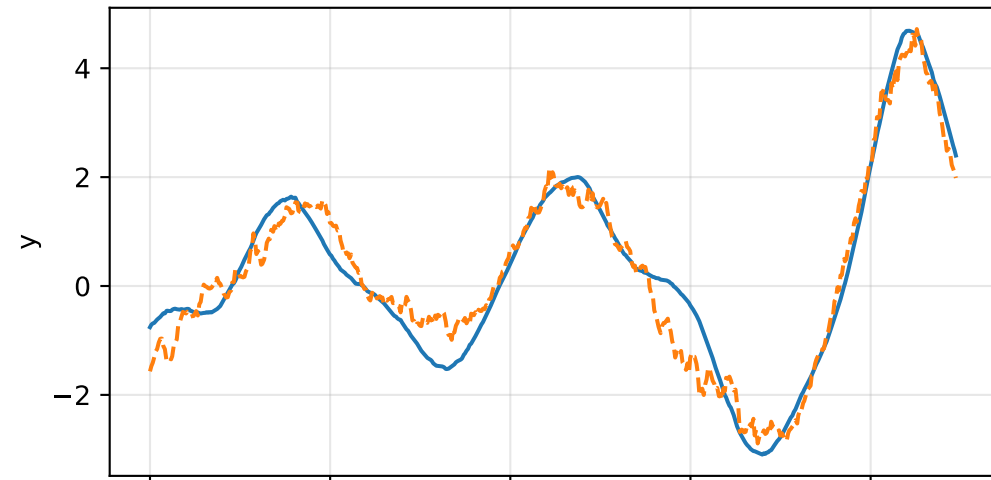
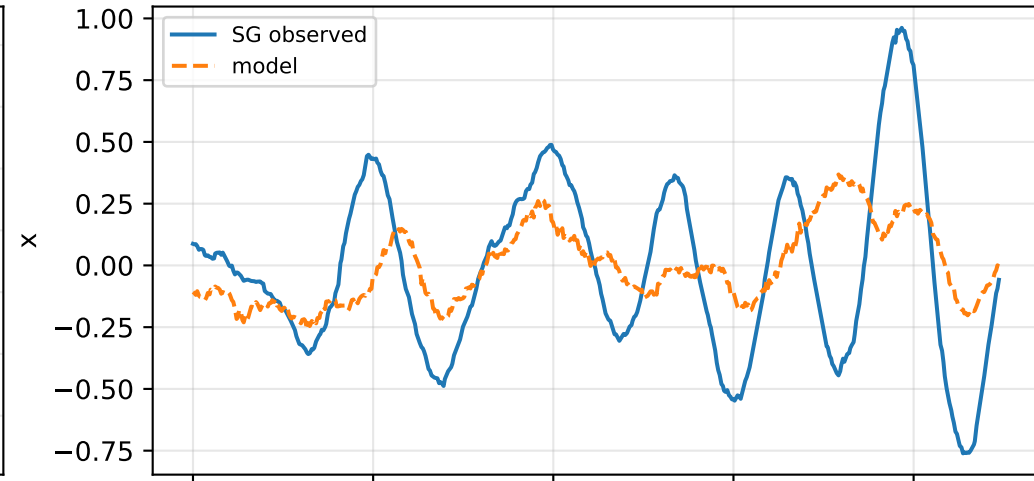


# cog\_z\_nominal: acceleration residual objective

## specific acceleration [m/s^2]

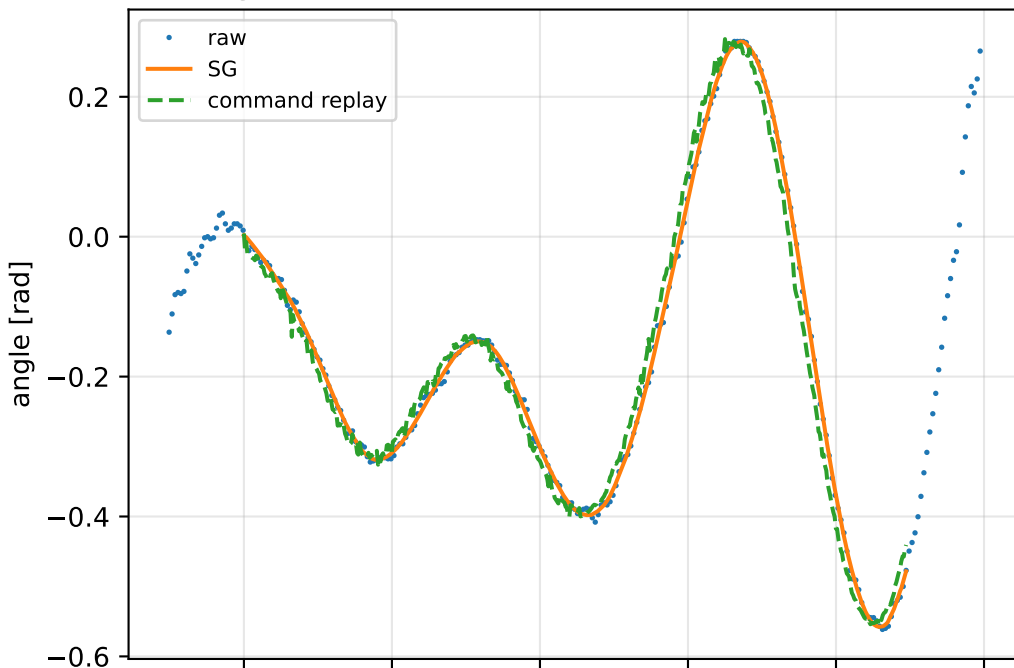


## angular acceleration [rad/s^2]

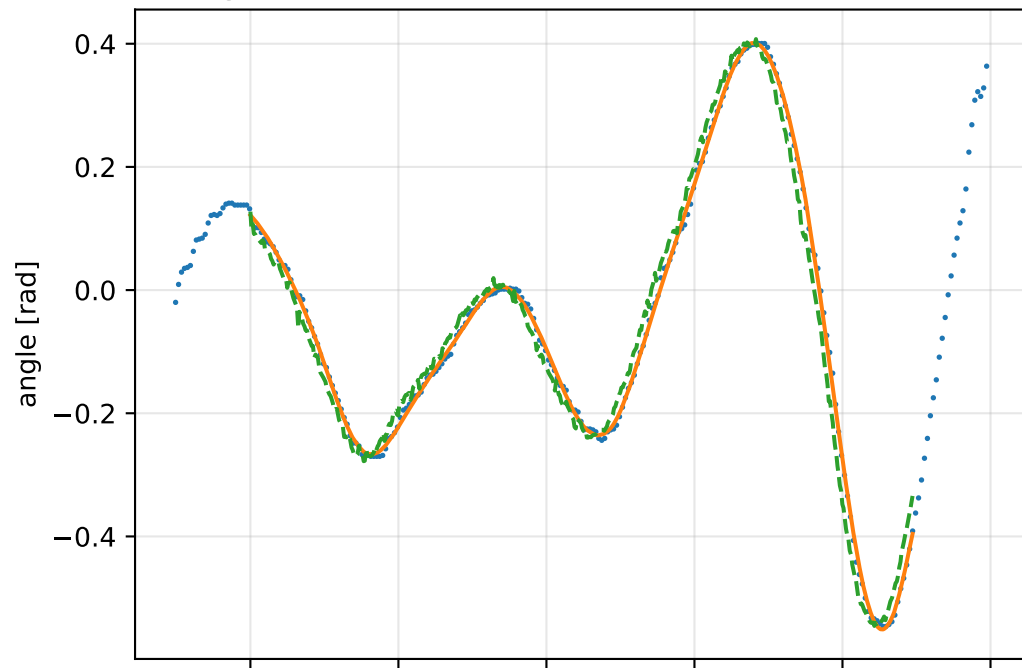


# cog\_z\_nominal: gimbal measurement audit (objective=measured\_sg)

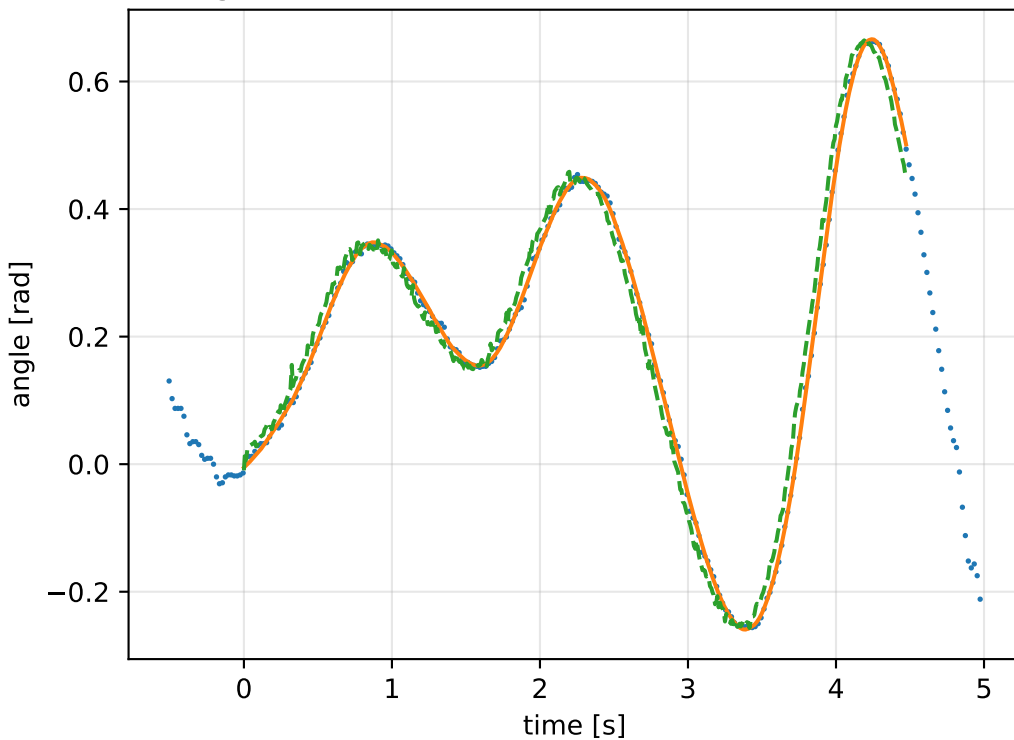
gimbal 1: RMSE=0.027, max=0.07529 rad



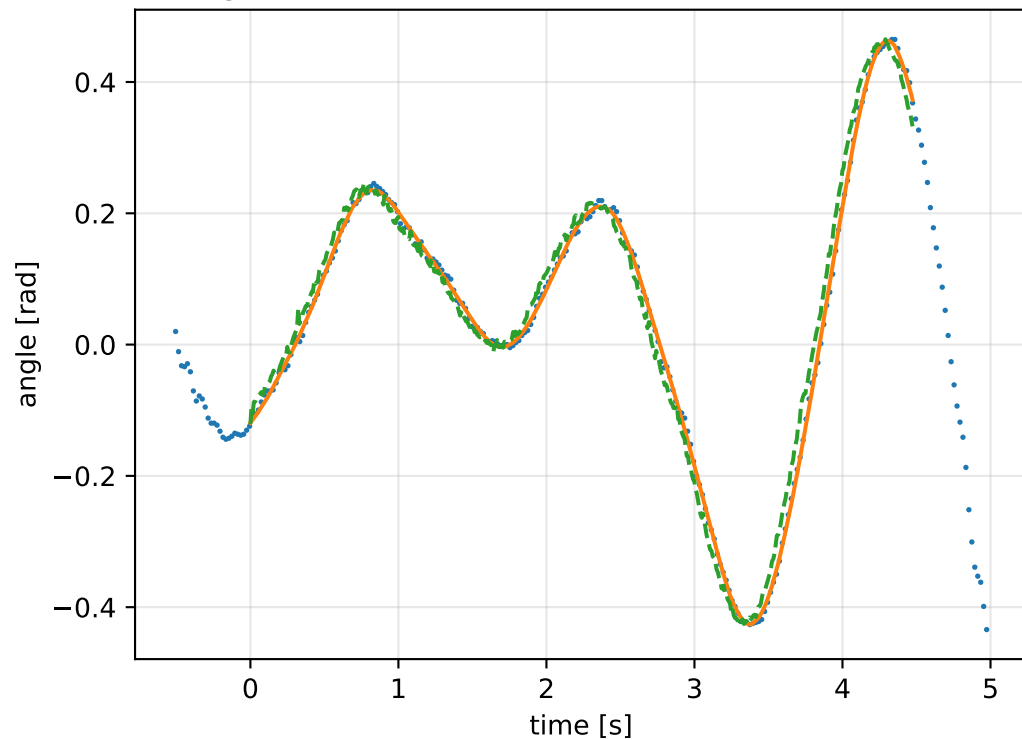
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

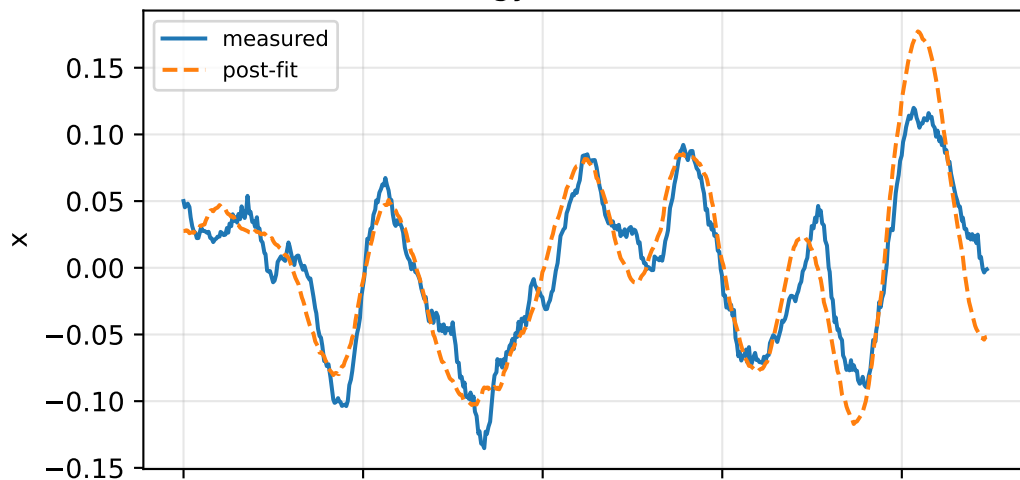


gimbal 4: RMSE=0.02598, max=0.07261 rad

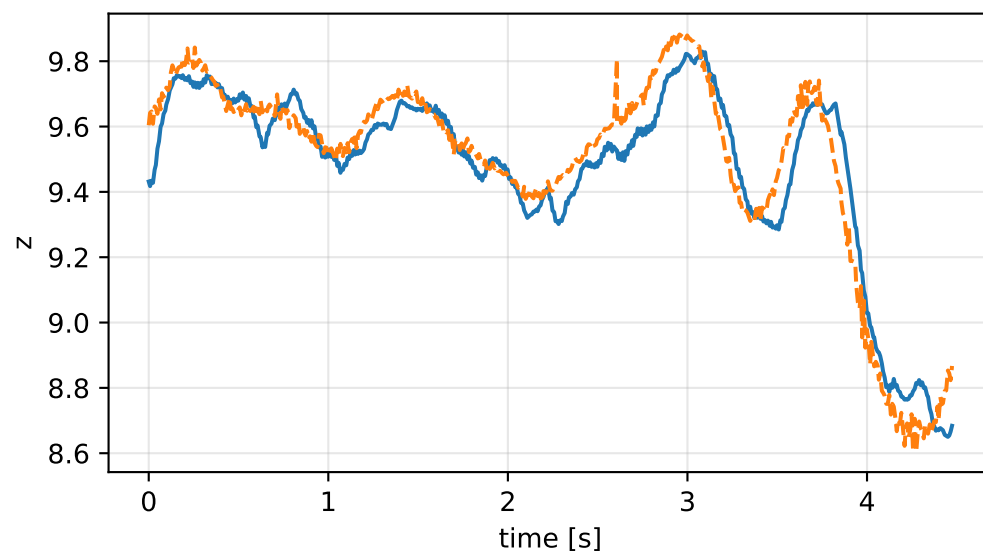
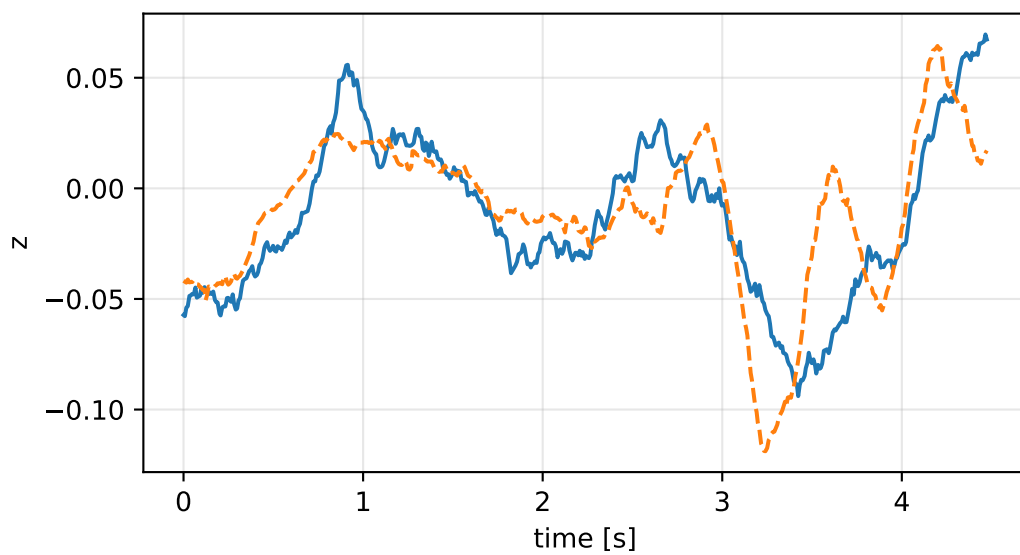
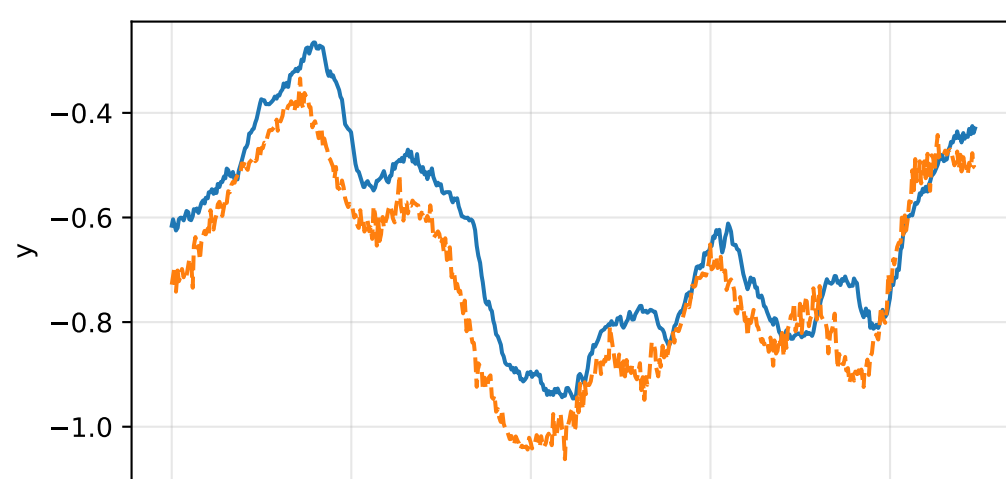
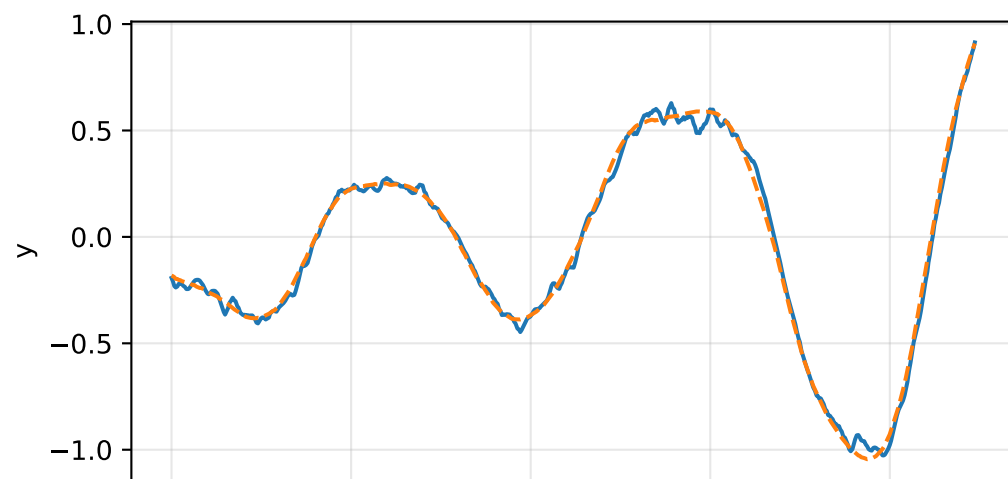
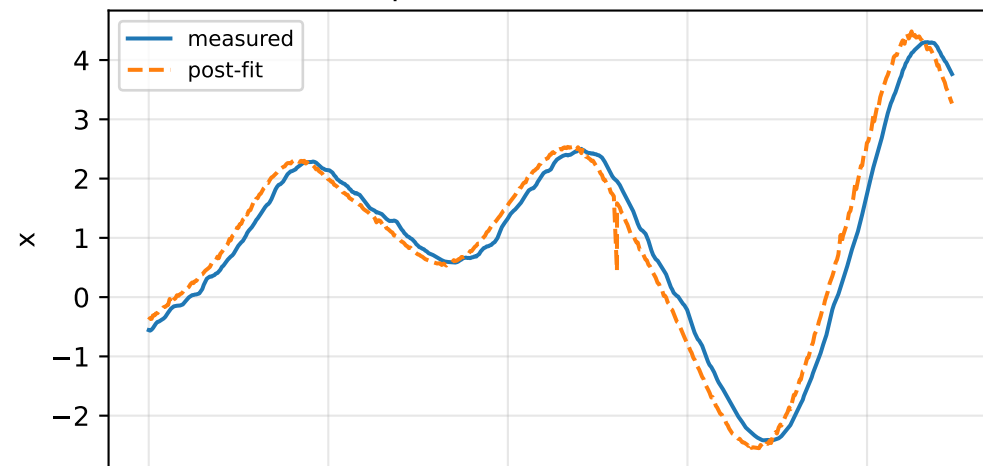


# cog\_z\_nominal: IMU comparison (diagnostic only)

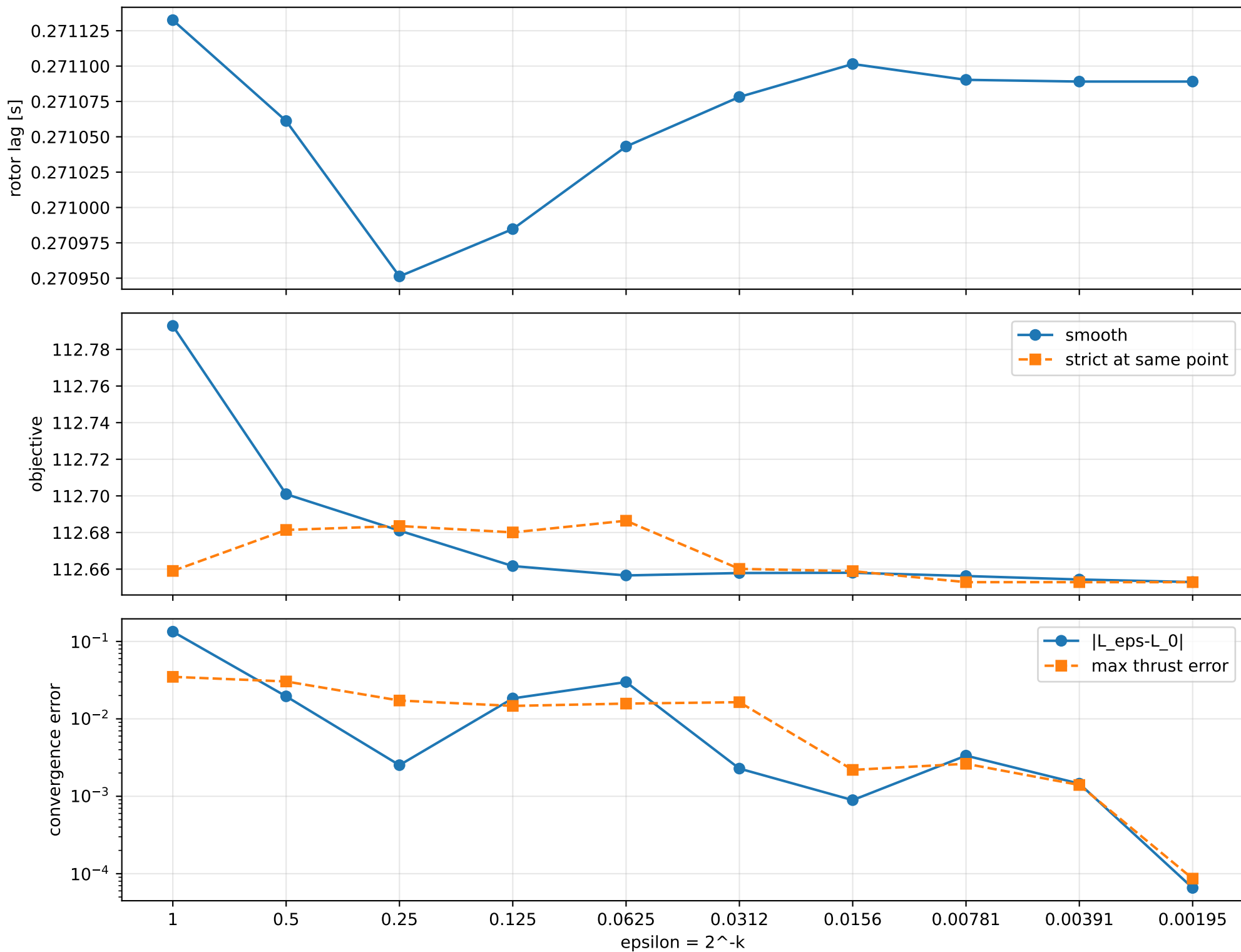
gyro [rad/s]



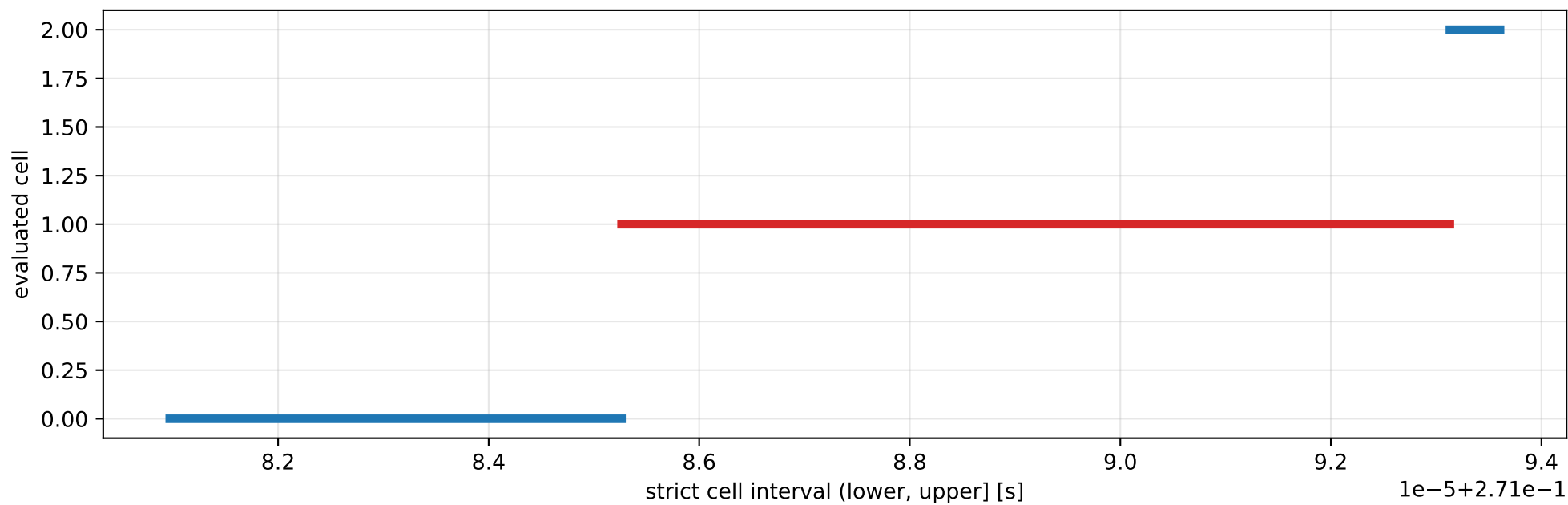
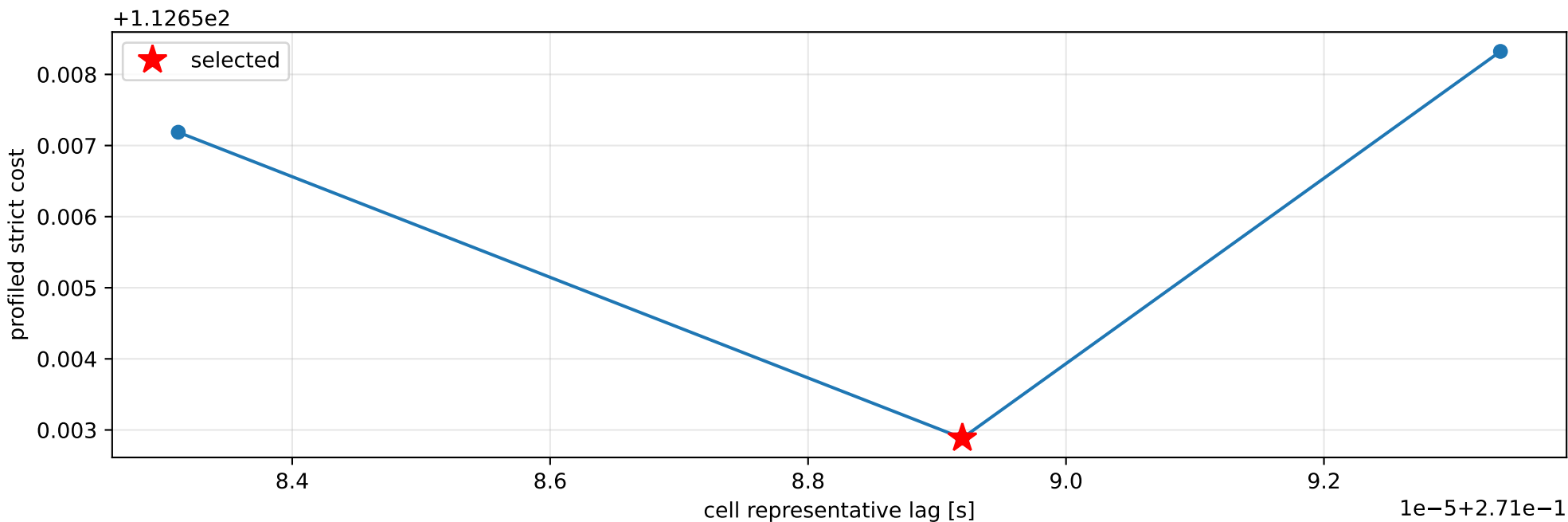
specific force [m/s<sup>2</sup>]



cog\_z\_nominal: smooth-to-strict continuation



# cog\_z\_nominal: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089068 s; data boundary=False

## cog\_z\_nominal: parameter estimate

Case: cog\_z\_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.29704142

inertia [kg m<sup>2</sup>]:

```
[[ 0.53737843  0.01333703 -0.00321896]
 [ 0.01333703  0.25328348 -0.07904027]
 [-0.00321896 -0.07904027  0.76665249]]
```

CoG body [m]: [-0.02636313 0.01128104 0.00950975]

force effectiveness: [2.78578832 2.25637994 2.48734099 2.98654304]

scale-free J/m [m<sup>2</sup>]:

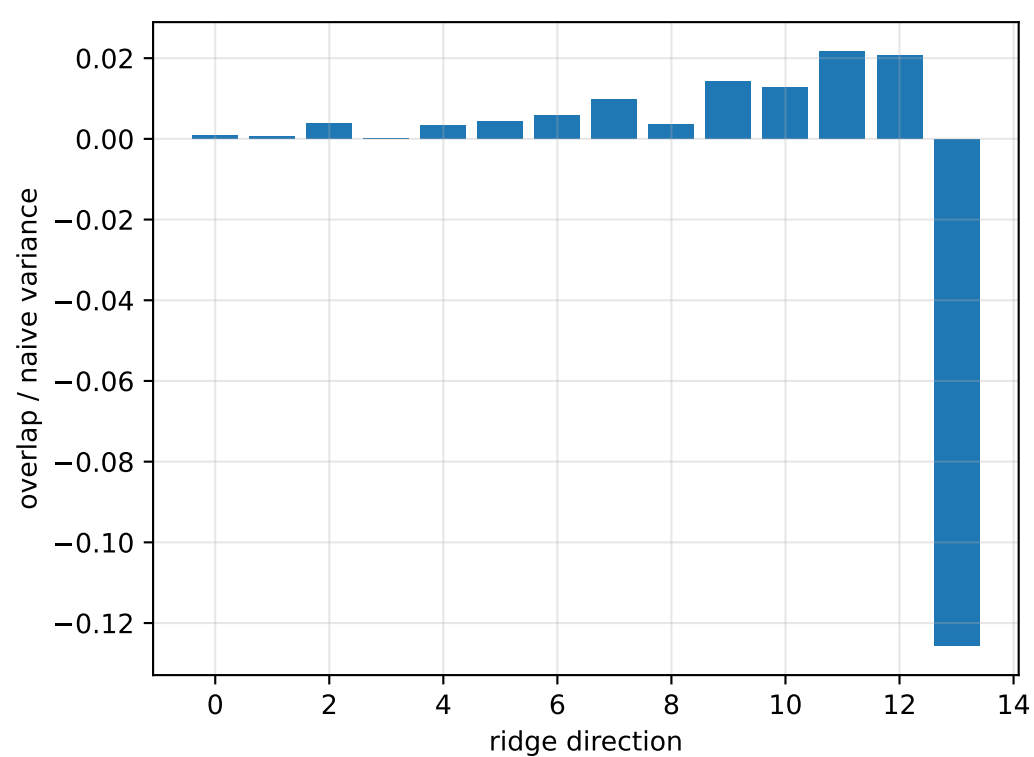
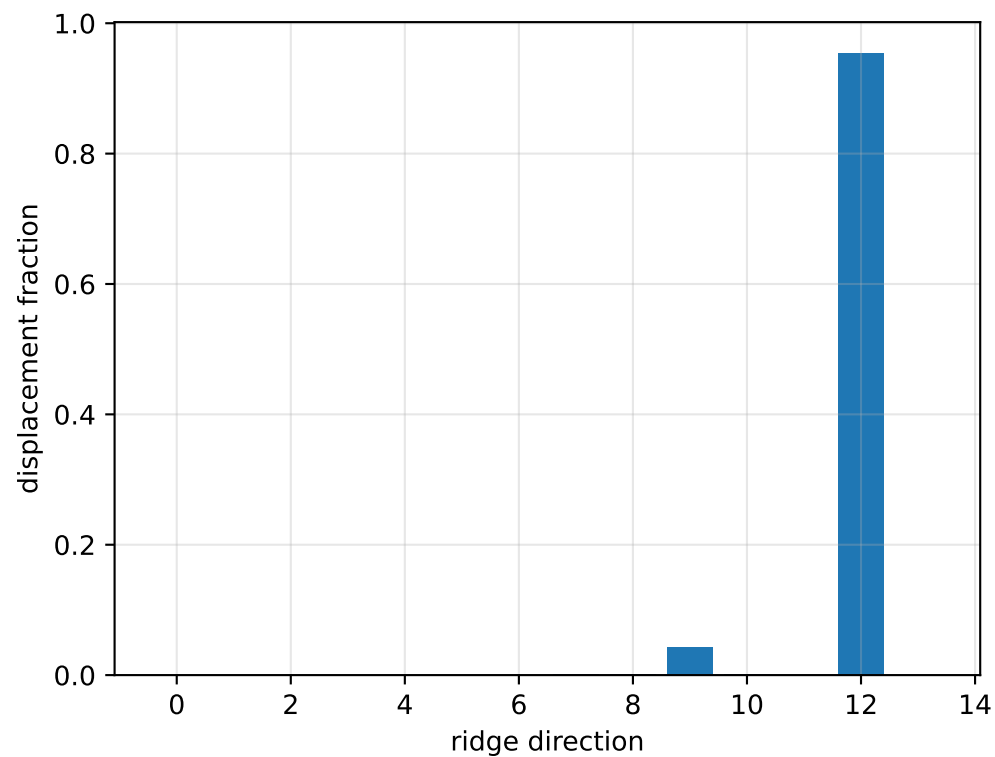
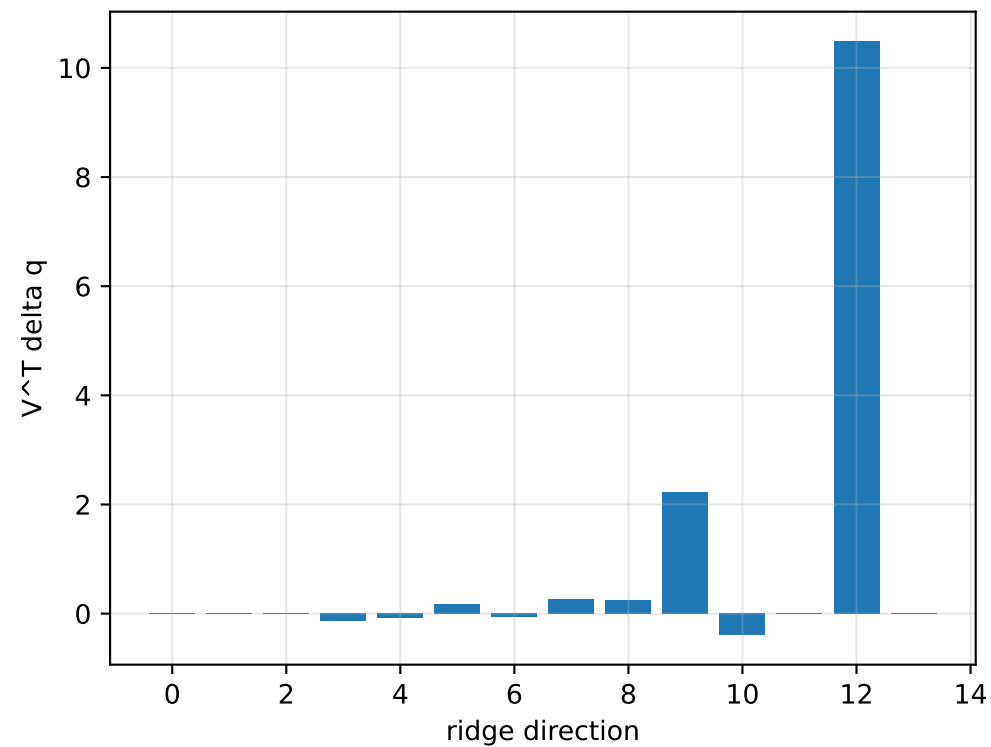
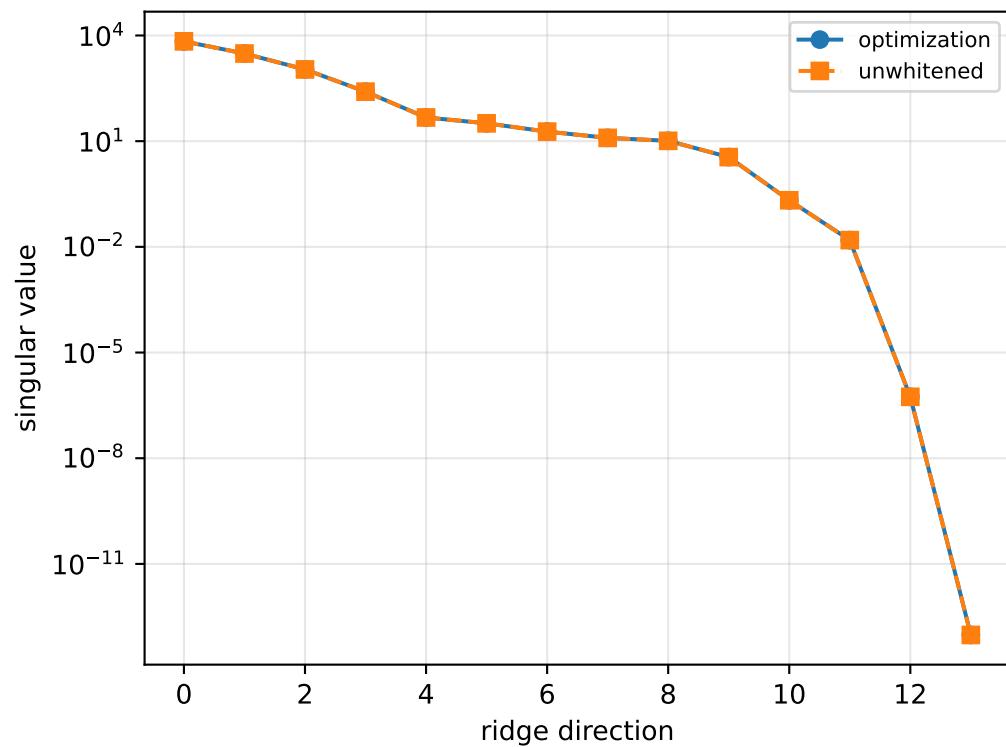
```
[[ 0.07364333  0.00182773 -0.00044113]
 [ 0.00182773  0.03471044 -0.01083182]
 [-0.00044113 -0.01083182  0.10506347]]
```

scale-free f/m: [0.38176956 0.30921846 0.34086979 0.40928136]

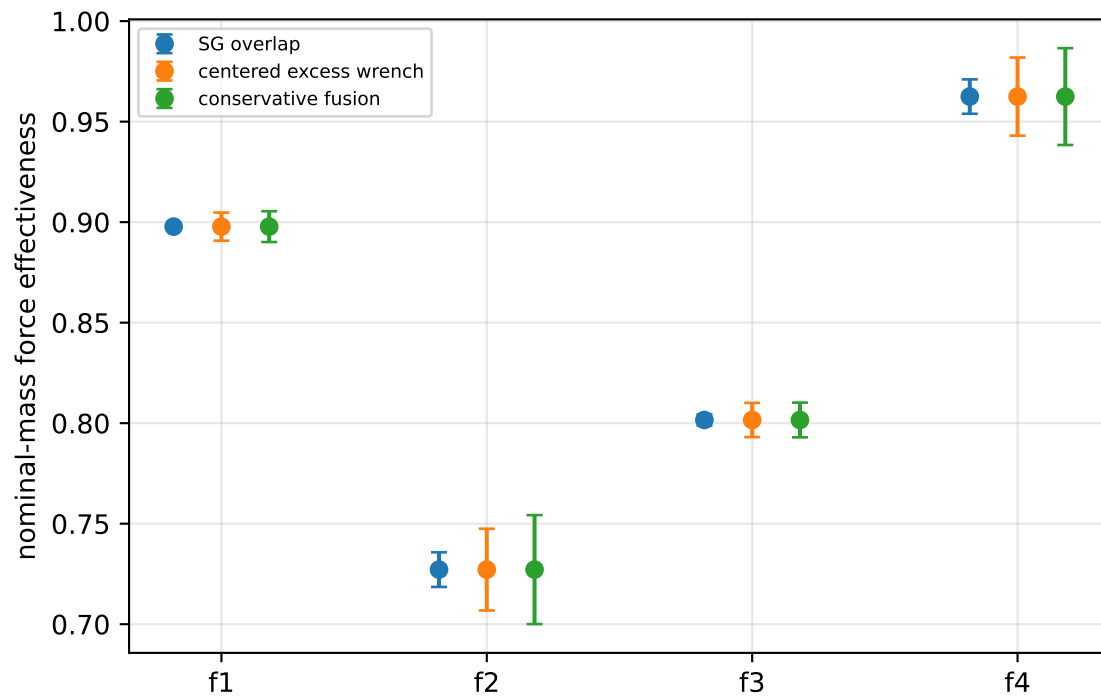
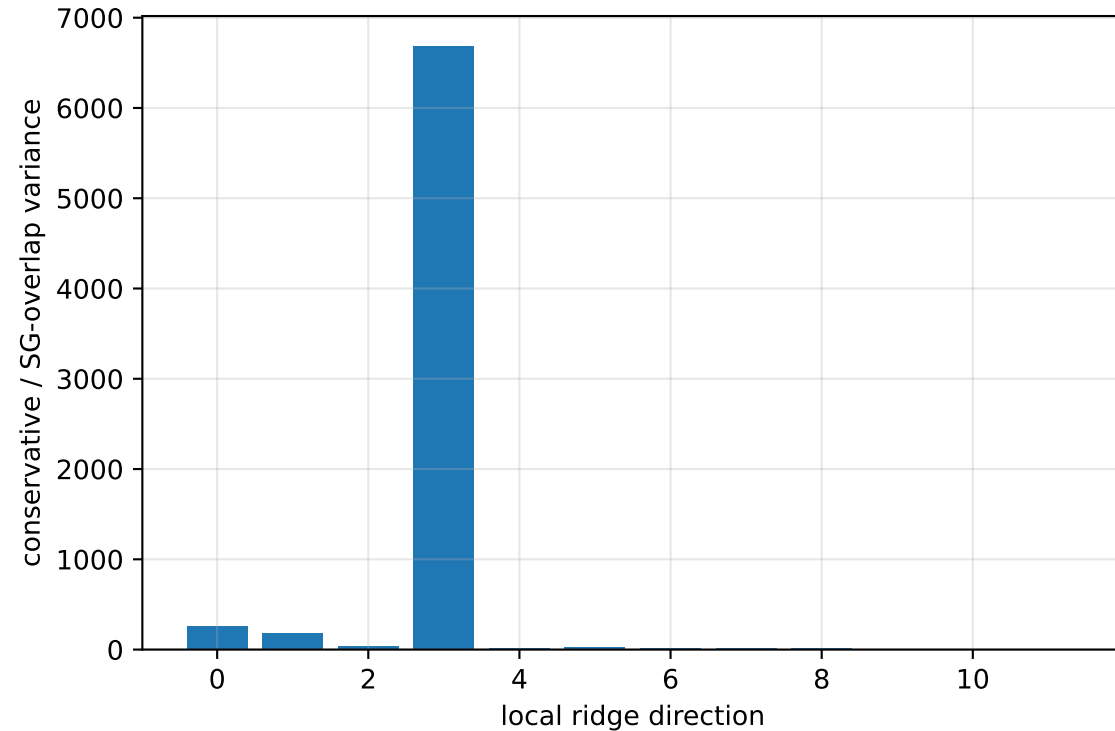
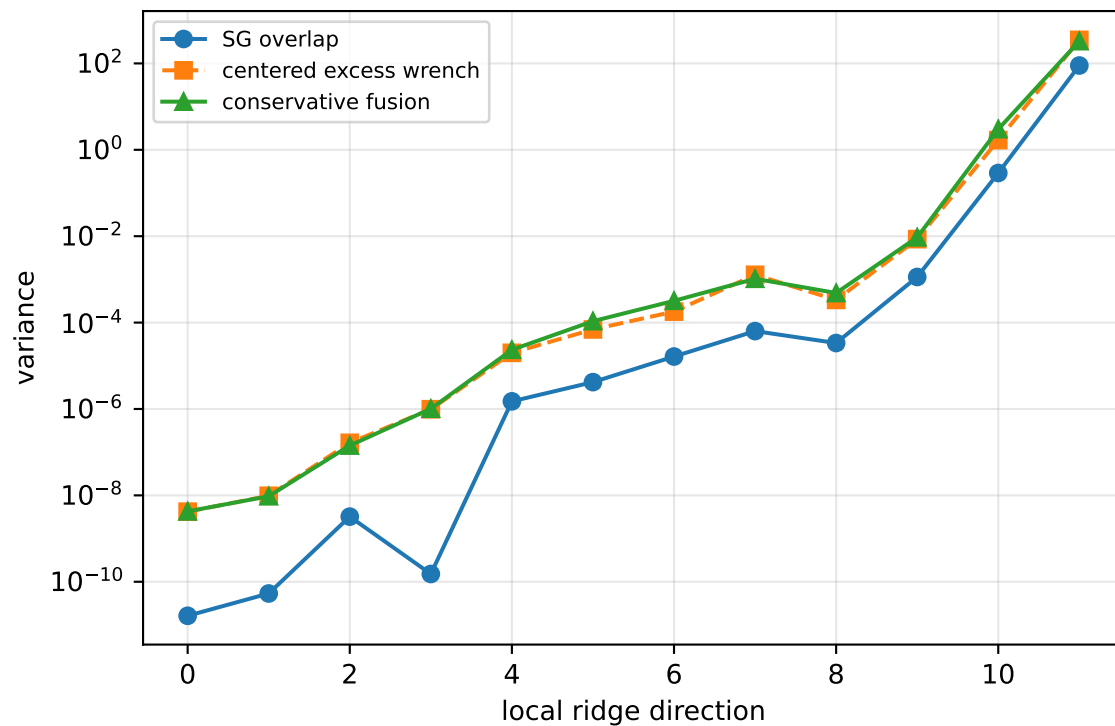
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

cog\_z\_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=1.5e-13)



## cog\_z\_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions  
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 229.4$   
wrench-to-acceleration recovery max error =  $2.83\text{e-}15$   
exact scale gauge ridge index = 13 (alignment 1)

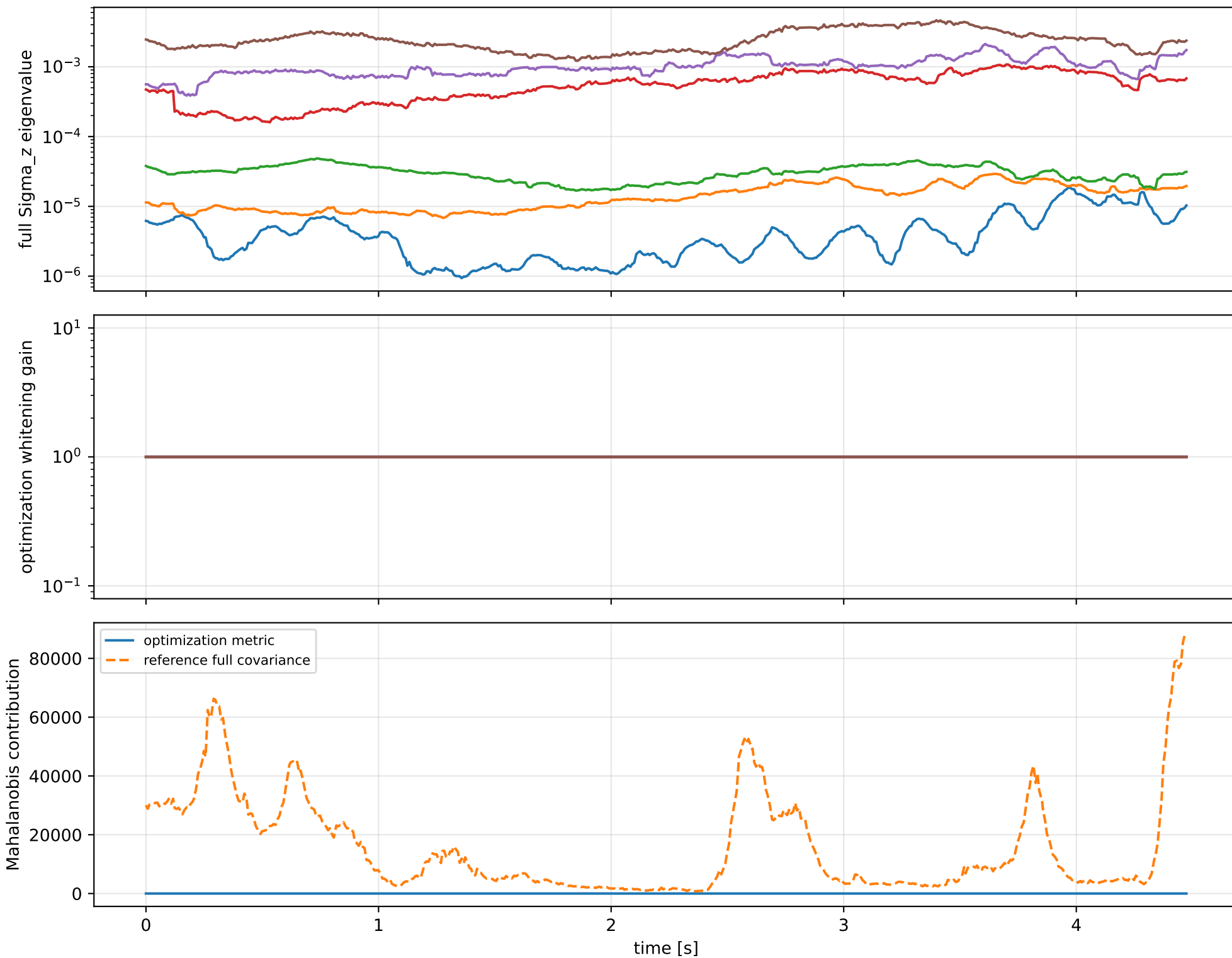
index 11: variance=327.6, ratio=3.664  
 $\log_{\text{mass}}=+0.00228$ ; second\_moment\_diag\_1=+0.165  
second\_moment\_diag\_2=-0.121; second\_moment\_diag\_3=-0.0548  
second\_moment\_offdiag\_12=+0.821; second\_moment\_offdiag\_13=-0.518  
second\_moment\_offdiag\_23=+0.116; cog\_x=-5.28e-05  
cog\_y=+3.14e-05; cog\_z=-7.3e-06  
log\_force\_eff\_1=+0.00236; log\_force\_eff\_2=+0.00183  
log\_force\_eff\_3=+0.00217; log\_force\_eff\_4=+0.0026

index 10: variance=3.006, ratio=10.35  
 $\log_{\text{mass}}=+0.0253$ ; second\_moment\_diag\_1=+0.0255  
second\_moment\_diag\_2=+0.56; second\_moment\_diag\_3=-0.716  
second\_moment\_offdiag\_12=+0.0345; second\_moment\_offdiag\_13=+0.0976  
second\_moment\_offdiag\_23=+0.399; cog\_x=+0.000956  
cog\_y=-0.00133; cog\_z=+0.000295  
log\_force\_eff\_1=+0.026; log\_force\_eff\_2=+0.0383  
log\_force\_eff\_3=+0.0228; log\_force\_eff\_4=+0.0175

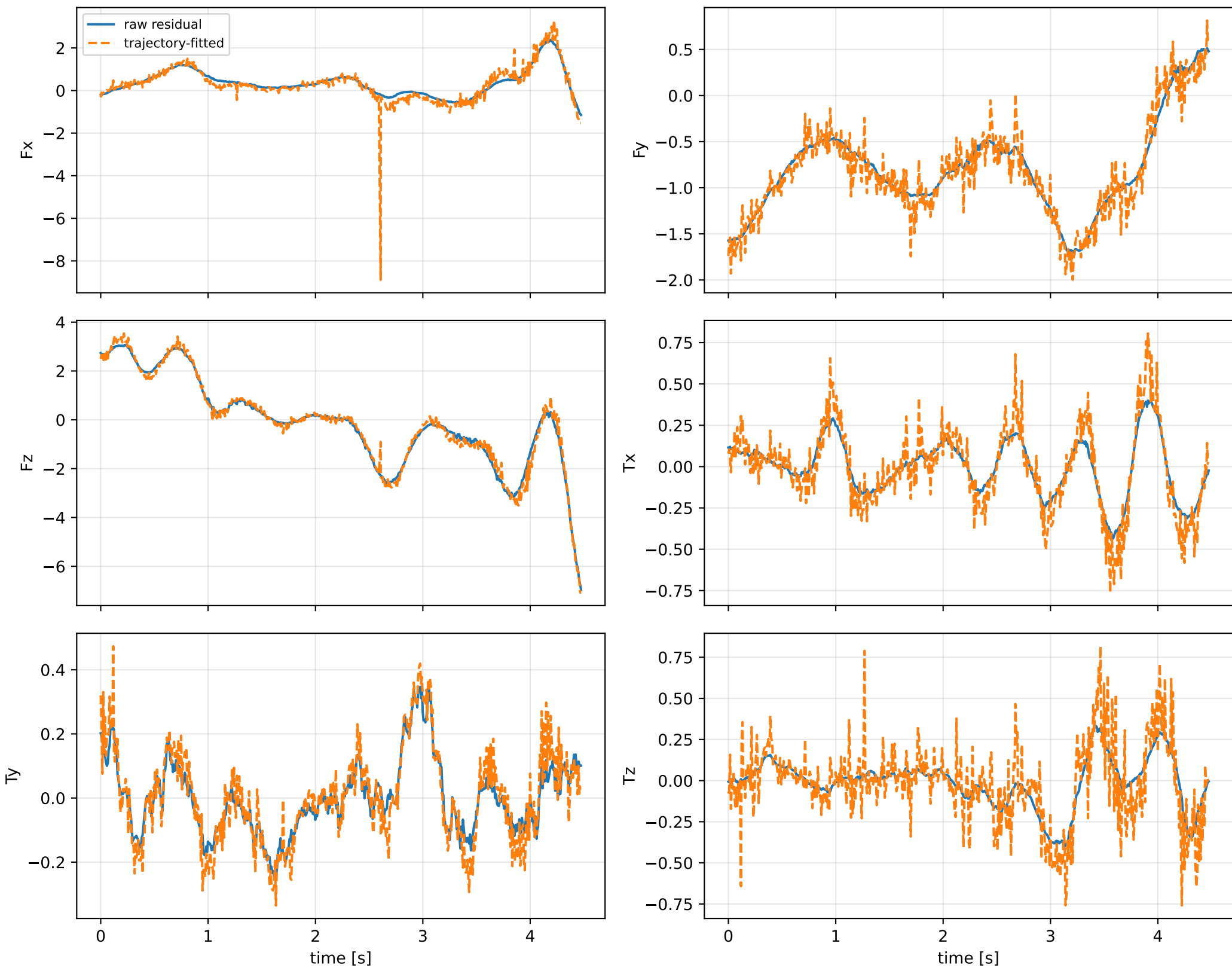
index 9: variance=0.009414, ratio=8.277  
 $\log_{\text{mass}}=-0.141$ ; second\_moment\_diag\_1=-0.132  
second\_moment\_diag\_2=+0.325; second\_moment\_diag\_3=+0.564  
second\_moment\_offdiag\_12=-0.00837; second\_moment\_offdiag\_13=-0.0519  
second\_moment\_offdiag\_23=+0.632; cog\_x=-0.0197  
cog\_y=+0.0106; cog\_z=-0.00276  
log\_force\_eff\_1=-0.104; log\_force\_eff\_2=-0.299  
log\_force\_eff\_3=-0.188; log\_force\_eff\_4=-0.0258

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog\_z\_nominal: optimization vs reference covariance



cog\_z\_nominal: wrench / closure (force max  $2.09\text{e-}14$ , torque max  $3.37\text{e-}16$ )



## cog\_z\_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo\_condition\_cog\_z\_nominal

role: pseudo\_conditioning\_ablation

source: /home/leus/catkin\_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo\_conditioning/scalar/cog\_z\_nominal.j

SHA256: 0fc7d4f0f59d87c49bab62593db16eefada27449488e6f94ba9dea5cb570cbdb

data objective: 112.652885396

prior objective: 6.23565823454e-12

total objective: 112.652885396

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog\_z\_nominal (cog\_position\_body\_m)

components: z

target: [0.00950975]

std: [1.e-05]

final: [0.00950975]

error: [-3.53147511e-11]

standardized residual: [-3.53147511e-06]

factor objective: 6.23565823454e-12